

# PLAYWRIGHT CAPSTONE PROJECT REPORT

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**Trainer:** Aryan Kumar

**Project:** IMDb Web Automation Testing Framework

**GitHub Repository:** [https://github.com/ToshRohan/Capstone-Project\\_Wipro\\_Rohan-Tosh](https://github.com/ToshRohan/Capstone-Project_Wipro_Rohan-Tosh)

## Executive Summary

This project presents an end-to-end test automation framework developed using **Playwright with JavaScript**. The framework is designed to validate multiple services of the **IMDb web application** through automated UI test cases, ensuring reliability, stability, and consistent user experience across different browsers.

The automation suite supports **cross-browser testing (Chromium, Firefox, and WebKit)** and is integrated with **GitHub Actions CI/CD pipeline** for continuous execution on every code push and pull request. The results are reported using **Allure Reports**, which provide detailed test execution insights including pass/fail status, logs, and execution history. The final reports are deployed using **GitHub Pages** for easy access and visualization.

The framework is built using the **Page Object Model (POM)** design pattern, which separates test logic from page-specific actions. This improves code reusability, scalability, and maintainability. Each IMDb service is structured as an independent module with dedicated test cases.

The project covers **seven IMDb services**, including modules such as Top 10 listings, Popular Celebrities, Movies/TV navigation, search functionality, and other key user-facing features.

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## Technology Stack

Playwright, JavaScript, Node.js, Git, GitHub, GitHub Actions, Allure Reports, GitHub Pages, VS Code.

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## Framework Architecture

The framework is based on the **Page Object Model (POM)** design pattern.

- Page classes handle locators and UI actions
  - Test files focus only on assertions and test flow
  - Utilities manage reusable functions and configurations
  - CI pipeline handles automated execution and reporting
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## Modules Covered

- IMDb Home & Navigation
- Search Functionality
- Top 10 / Top Rated Content
- Popular Celebrities
- Movie & TV Show Listings
- Individual Title Details Pages

- Footer & UI Components Testing

## GitHub Repository Structure

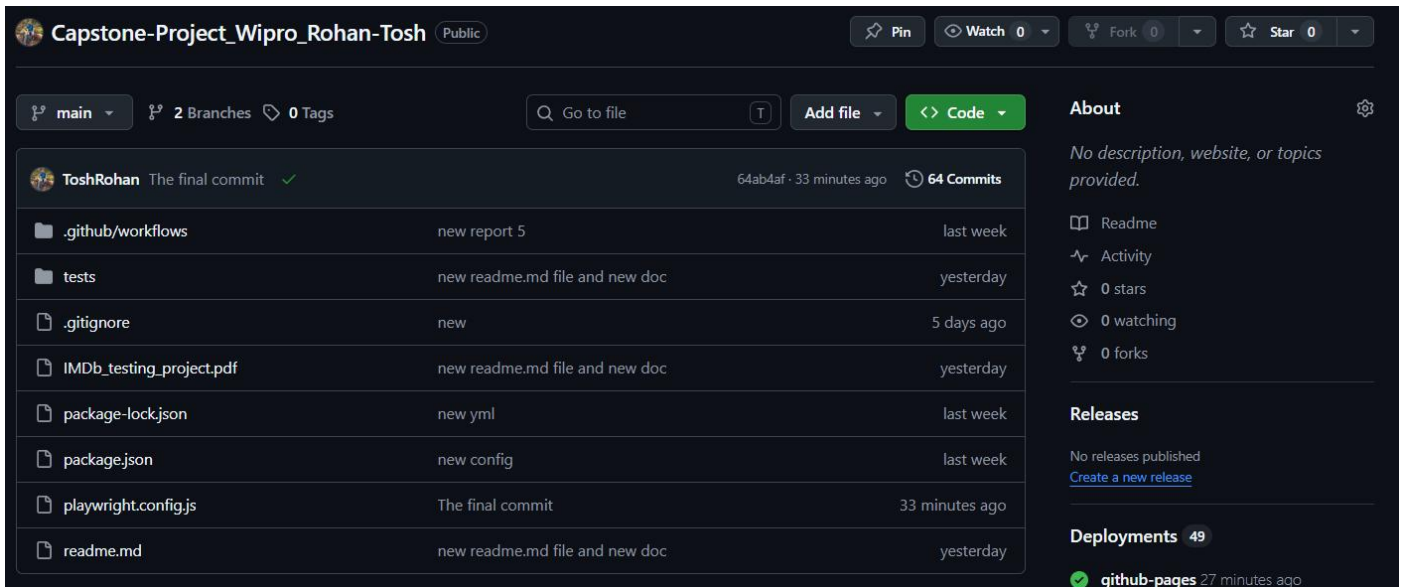


Figure: GitHub Repository Structure. This screenshot demonstrates the implementation and execution results of the Playwright framework.

## GitHub Actions Workflow Execution

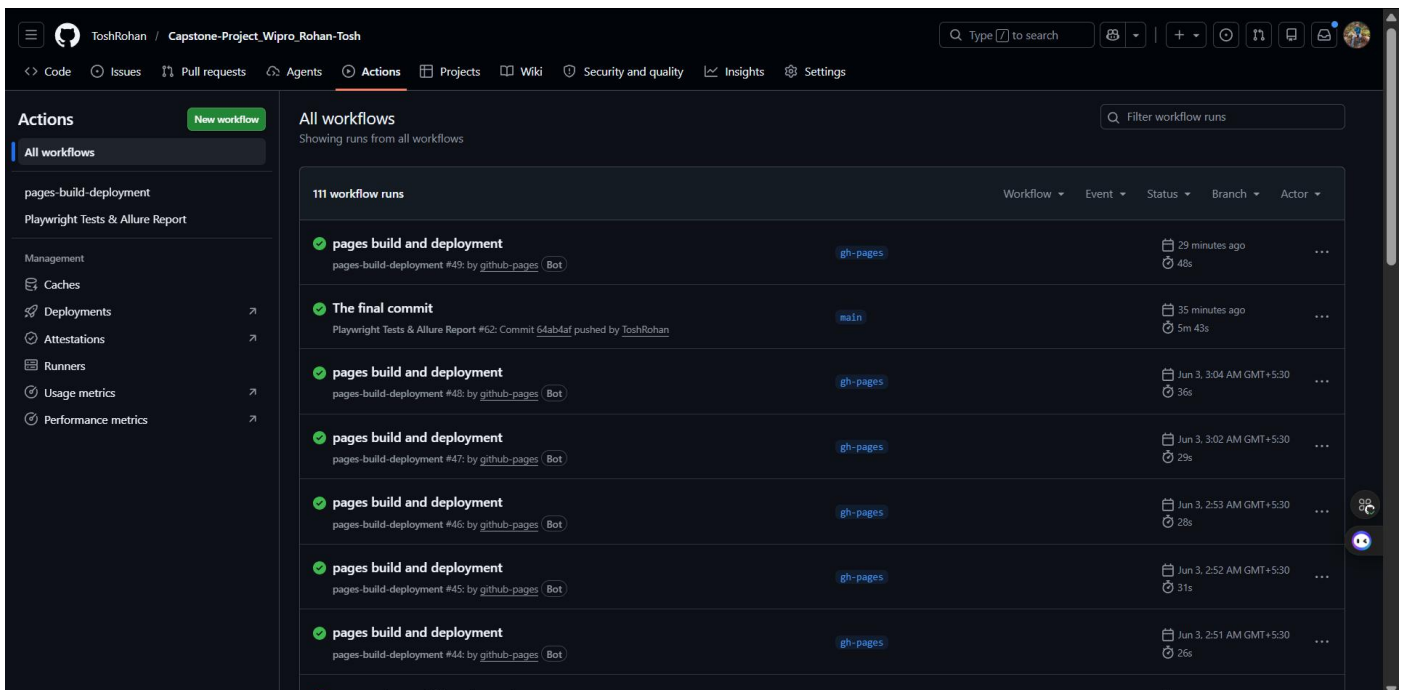


Figure: GitHub Actions Workflow Execution. This screenshot demonstrates the implementation and execution results of the Playwright framework.

## Allure Dashboard Overview

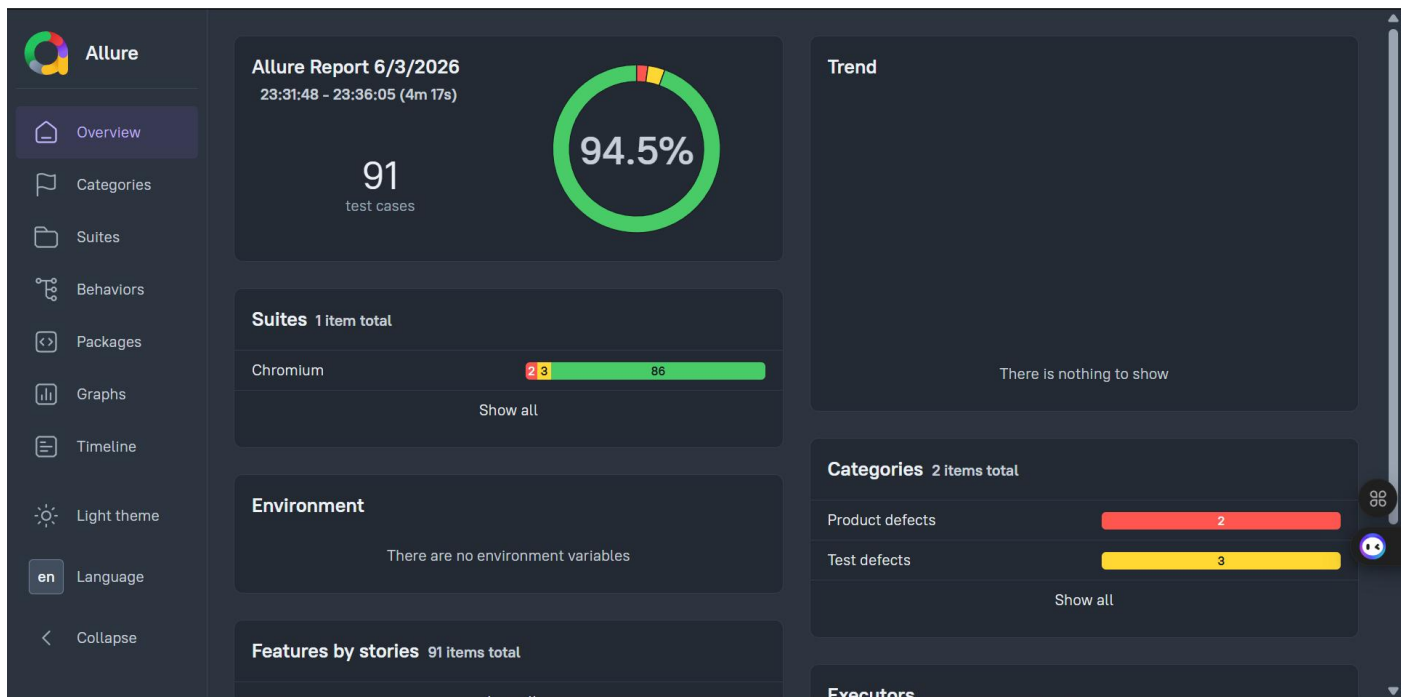


Figure: Allure Dashboard Overview. This screenshot demonstrates the implementation and execution results of the Playwright framework.

## Allure Graphs and Analytics

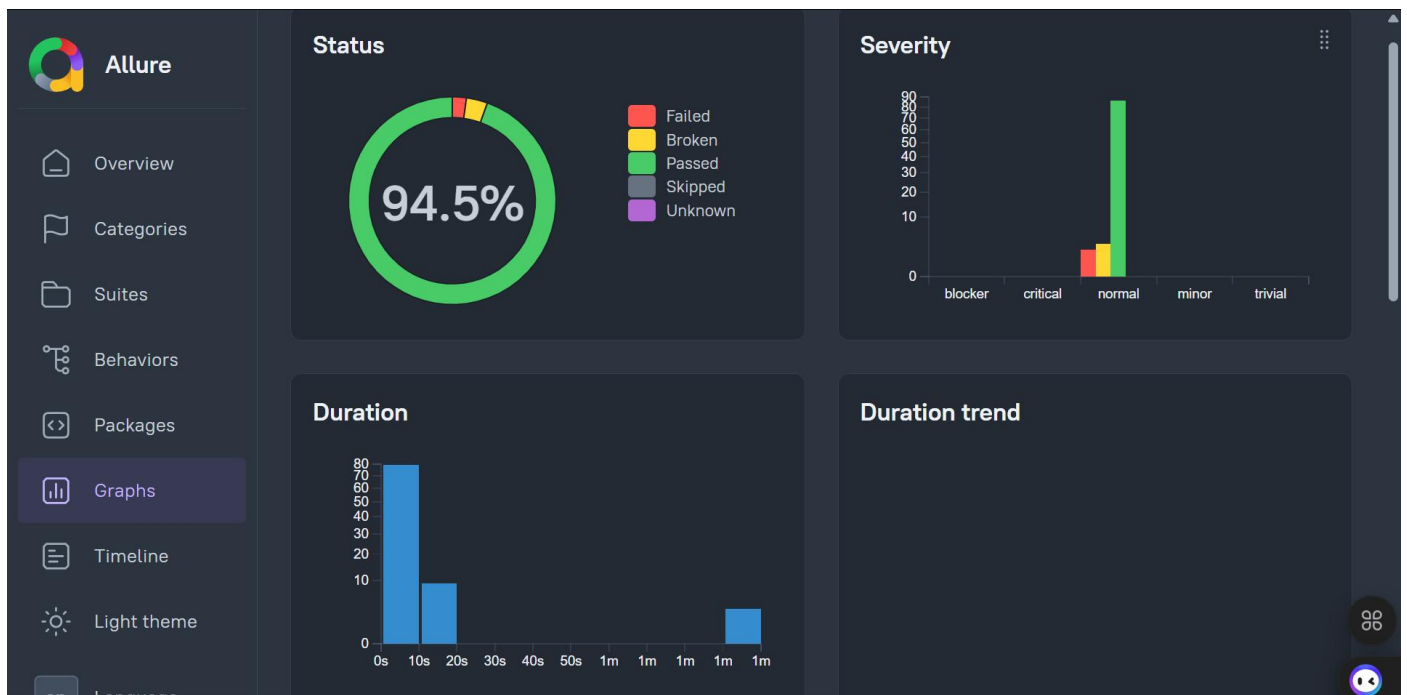


Figure: Allure Graphs and Analytics. This screenshot demonstrates the implementation and execution results of the Playwright framework.

## Local Allure Report Execution

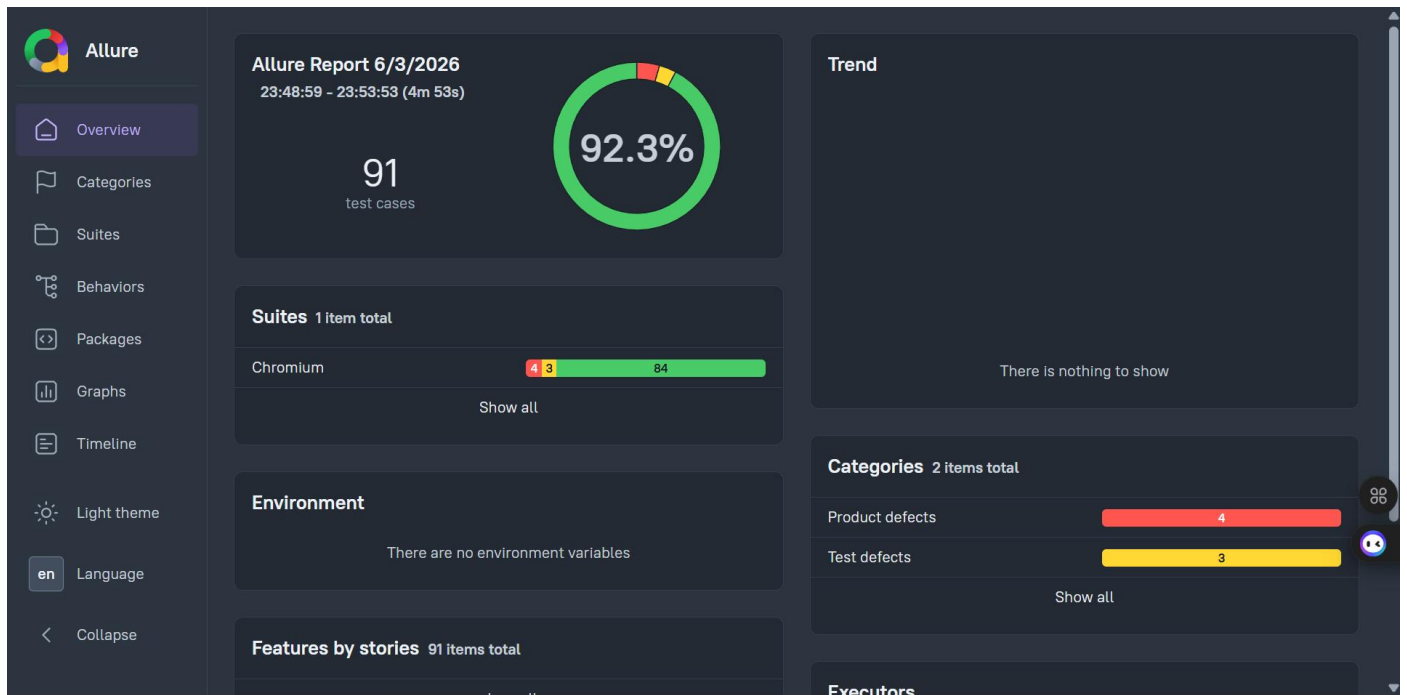


Figure: Local Allure Report Execution. This screenshot demonstrates the implementation and execution results of the Playwright framework.

### Execution Results

The test suite was executed on the **Chromium browser only**. Test reports were generated using **Allure** and successfully published via **GitHub Pages**. The **GitHub Actions CI/CD pipeline** handled automated execution of the test suite and ensured consistent report generation and deployment after each run.

### Key Achievements

- Automated UI testing using Playwright (Chromium browser)
- Implementation of Page Object Model (POM) architecture
- Basic test coverage for IMDb application services
- Allure reporting integration for execution tracking
- CI/CD pipeline setup using GitHub Actions
- Automated report generation and deployment via GitHub Pages

### Conclusion

This capstone project demonstrates a basic test automation framework using **Playwright with JavaScript**. The framework is structured using the Page Object Model and integrated with CI/CD and reporting tools. Currently, execution is limited to the Chromium browser, with scope for extending cross-browser support and scaling test coverage in future enhancements.